

ENGINEERING REBOOT 2026

Week 04 Master Workbook

Version 1.0

Week4 Baseline

Document Status

Document Type: Master Workbook

Classification: Planning Authority

Version: 1.0

Status: Approved and Locked

Authority: Week 4 Planning Review

Workbook Purpose

The Week 04 Master Workbook serves as the planning and curriculum authority for Days 16–20. Daily field manuals shall be derived from this workbook. This workbook establishes topics, exercises, projects, validation requirements, and deliverables for Week 04.

Week Theme

Files, Modules & Persistence

Week Learning Outcomes

- Persist information to storage.
- Load information from storage.
- Organize code across modules.

Day 16 – Text File Reading & Writing

Concepts

- open()
- read()
- write()
- append()
- with statement

Exercises

Filename: file_write_demo.py

Filename: file_read_demo.py

Project

Filename: personal_notes_manager.py

Validation Matrix

- File created successfully
- Data written successfully
- Data read successfully
- Notes persist after restart
- Program exits cleanly

Deliverables

- file_write_demo.py
- file_read_demo.py
- personal_notes_manager.py
- Notes.md
- Journal.md

Day 17 – File Processing & Data Analysis

Concepts

- Processing records
- Line-based processing
- Aggregation

Exercises

Filename: line_counter.py

Filename: word_counter.py

Project

Filename: expense_tracker.py

Validation Matrix

- Records loaded
- Totals calculated
- Summary displayed

Deliverables

- line_counter.py
- word_counter.py
- expense_tracker.py
- Notes.md
- Journal.md

Day 18 – CSV Fundamentals

Concepts

- CSV structure
- CSV reading
- CSV writing

Exercises

Filename: csv_read_demo.py

Filename: csv_write_demo.py

Project

Filename: student_grades_csv.py

Validation Matrix

- CSV read correctly
- CSV written correctly
- Records displayed correctly

Deliverables

- csv_read_demo.py
- csv_write_demo.py
- student_grades_csv.py
- Notes.md
- Journal.md

Day 19 – JSON Fundamentals

Concepts

- JSON structure
- Serialization
- Deserialization

Exercises

Filename: json_write_demo.py

Filename: json_read_demo.py

Project

Filename: inventory_json_manager.py

Validation Matrix

- JSON serialized
- JSON deserialized
- Data restored correctly

Deliverables

- json_write_demo.py
- json_read_demo.py
- inventory_json_manager.py
- Notes.md
- Journal.md

Day 20 – Modules & Imports

Concepts

- Module creation
- Imports
- Reusable code organization

Exercises

Filename: math_utils.py

Filename: import_demo.py

Project

Filename: calculator_module.py + main.py

Validation Matrix

- Module imports correctly
- Functions execute correctly
- Application separated across files

Deliverables

- math_utils.py
- import_demo.py
- calculator_module.py + main.py
- Notes.md
- Journal.md

Week Success Criteria

- Read text files
- Write text files
- Process stored records
- Read CSV files
- Write CSV files
- Read JSON files
- Write JSON files
- Create reusable modules
- Import functionality across files
- Complete Days 16–20

Week 05 Preview

- Exception Handling
- Input Validation
- Defensive Programming
- Type Hints
- Program Organization

Approval Record

Review Status: Approved

Workbook Status: Locked

Effective Beginning: Day 16

Publication Status: Approved for Archival

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